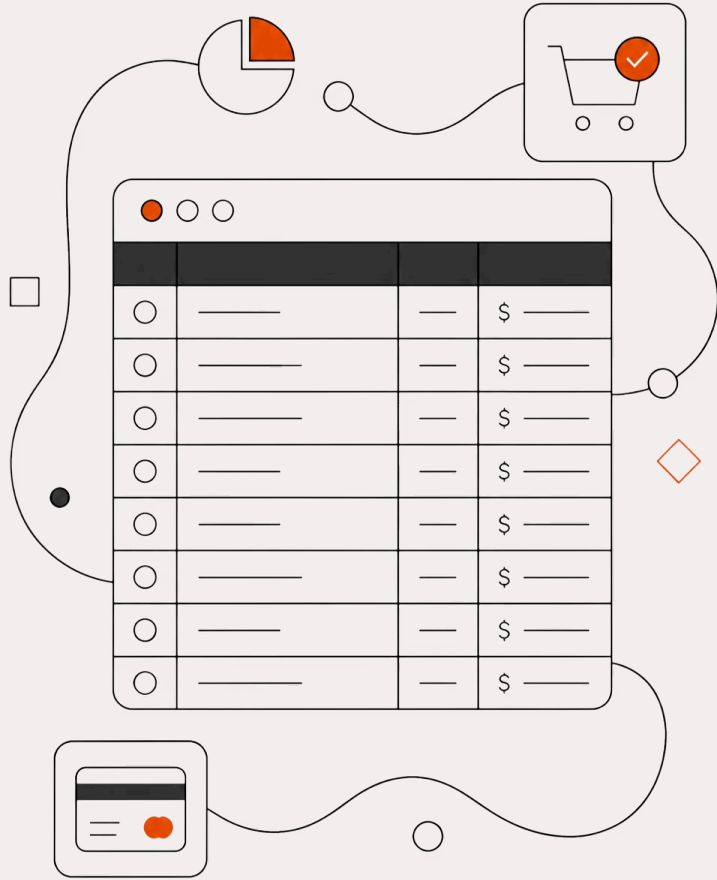




Customer Shopping Behavior Analysis

Analysis of 3,900 transactions to reveal spending patterns, segments, product preferences, and subscription behavior to guide strategy.



Dataset Summary

Scope

Rows: 3,900 · Columns: 18

Key Features

Demographics, purchase details, behavior signals (discounts, promo codes, frequency, ratings, shipping)

Data Quality

37 missing values in review rating column; handled with category median imputation.

Data Preparation & Feature Engineering

Loading with pandas, `df.info()` and `describe()` for structure and summary. Standardized column names to snake_case and documented changes.

Imputed missing review ratings using median per product category; removed redundant `promo_code_used` where `discount_applied` covered it.

New Features

age_group,
purchase_frequency_days,
customer_segment
(new/returning/loyal)

Consistency Checks

Verified discount vs promo consistency and payment method validity.



Profiling & Segments



Loyal

3,116 customers — high repeat purchases



New

83 customers — first-time buyers



Returning

701 customers — occasional repeat buyers

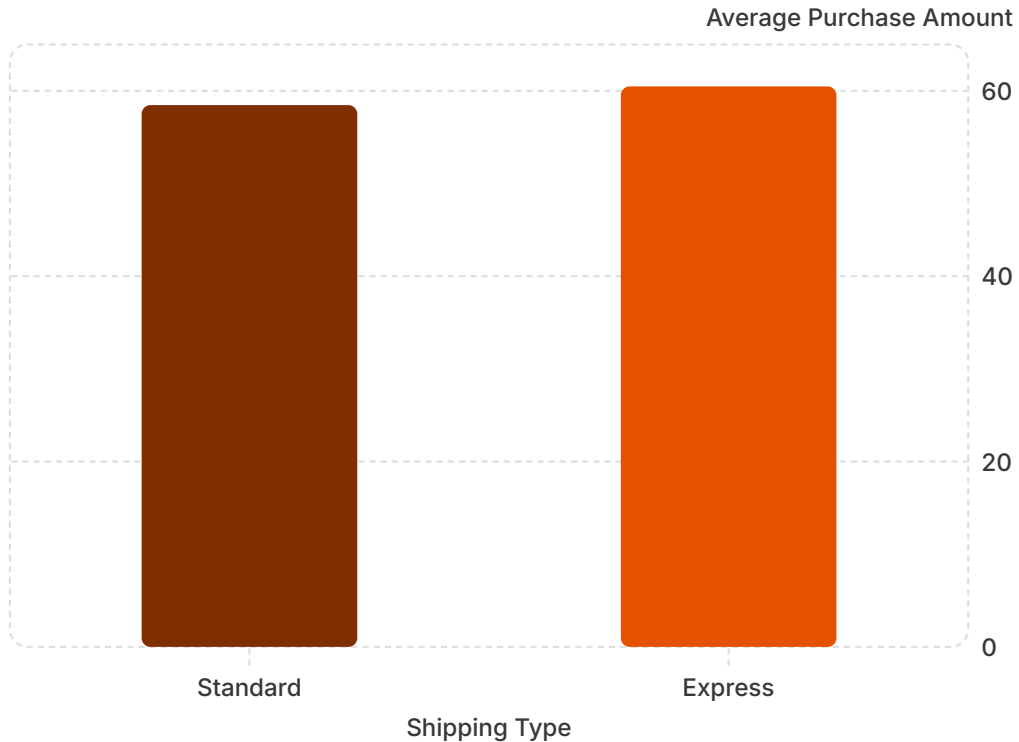
Revenue & Subscription Insights (SQL Results)

Revenue by gender: Female \$75,191 · Male \$157,890

Subscribers vs Non-subscribers — Avg purchase: No \$59.87 (Total \$170,436) · Yes \$59.49 (Total \$62,645)



Shipping & Spend Comparison



Customers using Express shipping spend slightly more on average than Standard.

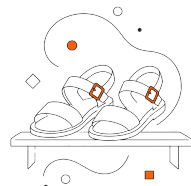
Use shipping insights to prioritize premium-shipping promotions for high-value items.

Top Products & Ratings



Gloves

Highest average rating — 3.86



Sandals

Average rating — 3.84



Boots

Average rating — 3.82



Hat

Average rating — 3.80

Also: Skirt among top 5 rated (3.79).

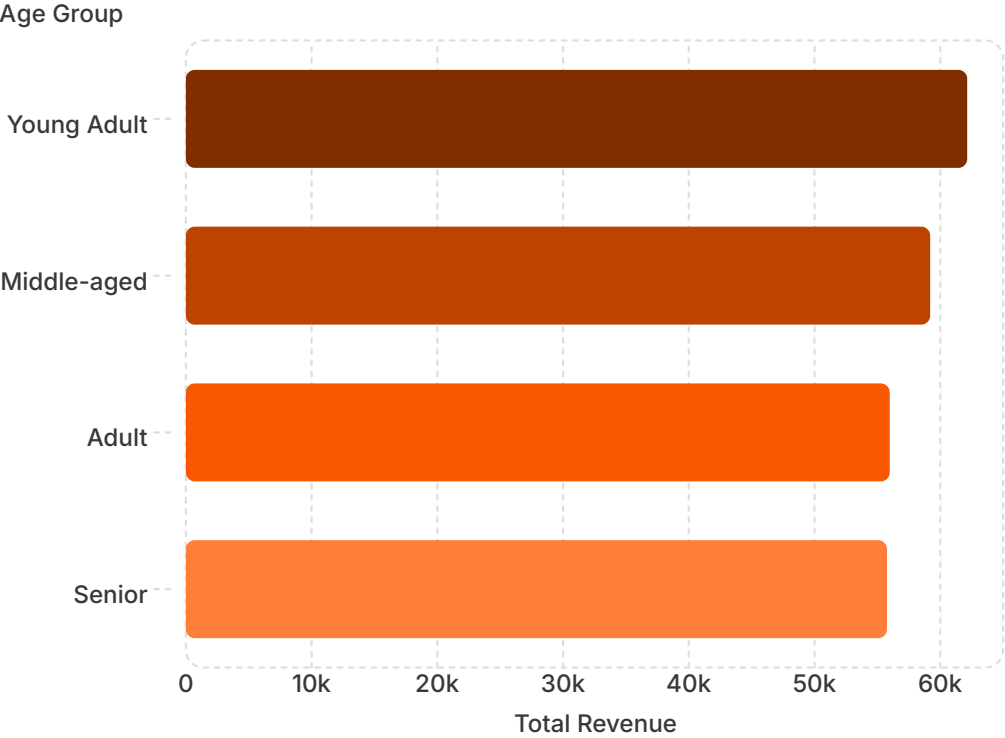
Discounts & Purchase Behavior

Top items by discount rate: Hat 50.00% · Sneakers 49.66% · Coat 49.07% · Sweater 48.17% · Pants 47.37%.

Identified high-spending customers who used discounts but spent above average — opportunities for margin optimization.



Category & Age Revenue Breakdowns



Top orders by category (sample): Clothing leads in total orders and revenue. Top items per category include Blouse, Pants, Shirt (Clothing); Sandals, Shoes, Sneakers (Footwear); Jacket, Coat (Outerwear); Jewelry, Sunglasses, Belt (Accessories).

Recommendations & Dashboard

Customer Behavior Dashboard (visual):



Key metrics shown: 3.9K customers · Avg purchase \$59.76 · Avg rating 3.75 · Subscription split 27% Yes / 73% No.

Boost Subscription

Promote exclusive benefits for subscribers to increase revenue and retention.

Loyalty Programs

Reward repeat buyers to move more customers into the Loyal segment.

Review Discount Policy

Balance discount-driven volume with margin protection for high-discount items.

Targeted Marketing

Focus campaigns on high-revenue age groups and express-shipping customers; highlight top-rated products.